



Aerial robot with 3D lidar permits safety at high speeds

Engineers and roboticists at the University of Hong Kong have developed SUPER, an aerial robot using 3D lidar



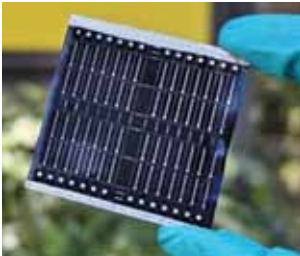
Safety-assured high-speed aerial robot (Credit: The University of Hong Kong)

technology to navigate unknown environments at high speeds while avoiding obstacles. Unlike traditional systems using cameras and sensors, SUPER's lidar provides real-time, 360-degree feedback

up to 70 metres, allowing the robot to generate optimal flight paths. Tested at 20m/s, SUPER excels in tracking targets and navigating through complex terrains, even in low-light conditions, with potential applications in law enforcement and military reconnaissance.

Hydrogen annealing increases efficiency of solar cells

Photovoltaic (PV) technology is advancing with the exploration of sustainable materials like $\text{Cu}_2\text{ZnSnS}_4$ (CZTS), a non-toxic, earth-abundant semiconductor. While CZTS



Hydrogen annealing increases efficiency of solar cells (Credit: <https://www.unsw.edu.au>)

offers a cost-effective, eco-friendly alternative to silicon, its efficiency has been limited to 11%. Researchers at the University of New South Wales (UNSW) improved this by using hydrogen annealing, which enhances carrier transport by redistributing oxygen and sodium, achieving a record 11.4% efficiency. The technique also shows promise for other materials like CIGS and aims to boost CZTS efficiency above 15%.

Solution found for electrical crosstalk in OLEDs having high pixel density

The rapid advancement of OLED technology, known for energy efficiency due to its lack of a backlight, faces challenges with electrical crosstalk at high pixel densities. Researchers from Hanyang, Yonsei, and Sogang Universities in South Korea have introduced a solution using a silicon-based small-molecule hole transport layer (SI-HTL) integrated via

microlithography. This innovation improves charge balance, enhances luminance, and reduces crosstalk, achieving a resolution of 10,062 PPI on a 15.24cm (6-inch) wafer, ideal for high-definition VR, AR, and wearable devices.

Chip-free wireless sensors cut waste

Engineers at the University of Glasgow have developed a wireless tagging system using cost-effective materials like silicon rubber (PDMS), carbon fibres, and coils, replacing



Chip-free wireless sensors (Credit: University of Glasgow)

traditional RFID tags with microchips. These chipless tags store data and monitor temperature in real time, which is ideal for food safety, health-care, and smart

clothing. Affordable, portable readers under £100 scan multiple tags simultaneously. With scalable manufacturing, this sustainable solution could reduce electronic waste and revolutionise supply chains by preventing food spoilage and enhancing medical safety.

Smart wheelchair with mixed control

Researchers at the University of Michigan have developed CoNav, a wheelchair controlled by a robot operating system (ROS) framework, combining manual control with autonomous navigation. Users can guide the chair via a joystick, while adaptive control blends their input with the chair's navigation plan. The system adapts to user direction, prioritising manual input when consistent. CoNav outperforms fully autonomous or manual wheelchairs in real-world tests.

AIoT capabilities with 'See Smarter' technology

Hikvision's AI-powered IoT (AIoT) integrates artificial intelligence with the Internet of Things, transforming security solutions. Its 'See Smarter' technology leverages deep learning algorithms for enhanced object detection, motion analysis, and perimeter protection. Key features include object classification, self-learning analytics, AcuSense technology, ANPR, and facial recognition. This system automates security processes, reduces costs, and improves response times. It also provides PPE detection, real-time people counting, and adaptive AI for diverse operational needs, advancing the AIoT field.